Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[T1]fontenc [utf8]inputenc [english]babel

[default,scale=0.95]opensans [scaled=0.85]beramono

listings tabularx booktabs grid-system
xcolor linkcolorrgb0, 0, 0.93
[hidelinks, colorlinks=true, urlcolor=linkcolor]hyperref
language=[LaTeX]tex,xleftmargin=2em

Grid System

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Grid system is a package that implements grid like layouts for LaTeX, as it is commonly known from CSS. You can easily divide your horizontal space into equal parts and assign these to boxes containing your content.

1 Usage

There are two environments to divide your area into boxes: `row` to divide your area into columns and `cell` to fill the area with content:

<Number of columns<Number of cells> <Number of columns to span> ... <Number of columns to span> ...
Each cell is created using a `minipage` environment. In future versions the will be a switch to choose either minipages or parboxes.

2 Parameters

There is one optional parameter for the environment `row` right now:

llX**Parameter Default Description**,
cellsep 1.75em horizonal space between two cells.

3 Contribute

You want to contribute? Just fork me on Github, start discussions and send me you pull requests:

`bitzl/latex-grid-system`

4 License

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5 Example

[cellsep=0.75cm]33 1

Overview

This shows what you can do with the `grid-system` package and three columns. 1

Why use it?

Sometimes you need to split your text in several independent columns based on equal division of the available space. This package allows you to easily do so. 1

Contribute

You want to contribute? Just fork me on Github, start discussions and send me you pull requests:

`bitzl/latex-grid-system`

[cellsep=0.75cm]32 2

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Why use it?

Sometimes you need to split your text in several independent columns based on equal division of the available space. This package allows you to easily do so.

[cellsep=0.75cm]32 1

Short

This shows the capabilities of grid-system with three equal columns. 2

Long

Sometimes you need to split your text in several independent columns based on equal division of the available space. This package allows you to easily do so.